

Saghil Abhayadev

Power Electronics Engineer

CONTACT

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📍 Bangalore, India

🌐 Saghil Abhayadev

EDUCATION

M.TECH IN POWER SYSTEMS from
Government Engineering college
Thrissur. **2019–2021**

◇ CGPA-9.57

B.TECH IN ELECTRICAL & ELECTRONICS
ENGINEERING from
Government Engineering college
Thrissur. **2015–2019**

◇ CGPA-7.86

CLASS 12 from
Vivekodayam BHSS Thrissur **2013–2015**
◇ Percentage-96%.

CLASS 10 from
Assisi EMHS Thalakkottukara. **–2013**
◇ Percentage-90%.

PERSONAL INFORMATION

Gender: Male

Marital status: Single

Languages known: English, Malayalam,
Kannada

SKILLS

- DFMEA & Derating analysis
- EMI-EMC std understanding
- Proficiency in LtSpice, Matlab, Simetrix simulation
- Debugging and trouble shooting
- Soldering
- MS Word, Excel, PowerPoint

SOFT SKILLS

- Proactive and self driven.
- Strong with oral and written communication
- Ownership mindset
- Work in close co-ordination with team and quality delivery

HOBBIES

- Karate, Running, football, movies

SUMMARY

Passionate & experienced power electronics hardware engineer with expertise in design, analysis, and validation of power electronic converters. Strong background in DC-DC, AC-DC power supplies. Proficient in simulation, prototyping, and testing of power systems with a focus on efficiency, reliability, and performance optimization. Experience in wide-bandgap semiconductors (GaN/SiC) devices.

EXPERIENCE

HARDWARE ENGINEER at *Ohmium Operation India Pvt. Ltd.*

Aug 2023–Present

- Project 1: 15kW DC-DC power supply
 - Worked on a 15kW half bridge DC-DC CC/CV power supply for PEM electrolyzer application.
 - Worked on designing transformer using planar transformer technology for compact design.
 - Worked on designing of a input EMI filter board.
 - Performed detailed loss calculation and simulation studies for optimizing the converter performance.
 - Performed closed loop tuning for improving dynamic responses.
 - Contributed to thermal management by optimizing the component sizing and airflow.
- Project 2: T-Type NPC AC-DC converter
 - Developed a gate driver power supply with isolated output using push-pull converter topology.

TEST VALIDATION ENGINEER at *Nexgen Power System Pvt. Ltd.*

Aug 2022–July 2023

- Performed switching characterization of vertical GaN devices using double-pulse testing (DPT).
- Quantified switching losses (Eon, Eoff) and timing parameters (turn-on/off delays, rise/fall times) across various operating conditions.
- Led testing and validation of a 240W AC-DC laptop charger.
- Performed Board bring-up and RCA of failed units in R&D.

GRADUATE ENGINEER TRAINEE at *enArka India Pvt. Ltd.*

Nov 2021–July 2022

- Testing and validation of 30kW DC-DC converter and 60kW AC-DC converter.
- Performed reliability analysis of converter using Windchill PTC software and estimated MTBF for individual PCBAs.
- Executed test plan and test setup along with documentation of test results and report making.

ACHIEVEMENTS

- Qualified Gate Exam in the year 2019 and 2021.
- Published 3 conference papers in IEEE related to the academic project.

ACKNOWLEDGMENT

I hereby confirm that all the information included in my resume is true and accurate to the best of my knowledge. I take the full responsibility for the authenticity of the details mentioned.

Saghil Abhayadev